

# Beyond the Single Request: Detecting Multi-Turn Attacks in the Model Context Protocol

**Abstract**—Almost all security defenses framed for Model Context Protocol (MCP) evaluate each request independently: a policy function of the form  $f(\text{request}, \text{config}) \rightarrow \{\text{allow}, \text{block}\}$ . We show this class of evaluator is blind to a documented attack pattern — multi-step sequences in which no individual call satisfies any single-request blocking predicate, yet the accumulated sequence constitutes an exploit. We formalize this as a protocol-layer necessity argument: single-turn evaluation fundamentally cannot detect state-dependent attack sequences, regardless of how per-request rules are tuned. Existing multi-turn defenses address this gap at the application and LLM-context layers, relying on expensive guard-model inference to scan semantic context across turns. We demonstrate that such overhead is unnecessary: keeping track of session state locally on each connection/worker handling the request is enough and a minimum to stop single-source attacks. Doing this sequence tracking adds a tiny delay of just 0.008 ms — 100,000 times faster than calling an LLM/guard model — bringing the total proxy pipeline delay to just 0.153 ms.

We present MCP-Shield, a protocol-layer proxy operationalizing the ATTESTMCP mitigations extended with session-level call-history tracking and sliding-window sequence detection. Across 10 sequence-dependent attacks, single-turn evaluation achieves a 0.0% detection rate while stateful tracking achieves 100.0%. We map the operational boundary across sliding window sizes  $w \in \{2, 3, 4, 5, 7, 10\}$  and demonstrate that while false positives spike to 13 of 14 (carefully engineered cases to flag a false positive) at  $w = 4$ . In a cross-server multi-turn attack, an attacker splits a malicious sequence across multiple backend servers — sending step 1 to Server A, step 2 to Server B, and step 3 to Server C — so that no single server’s session history contains the full sequence. We refer to this as a cross-server multi-turn attack. When servers operate in isolation, no single server sees the full context, allowing the attack to succeed. To solve this, we tested pooling all user activity into a centralized session system (aka Client-Unified Tracking Topology). In a standalone simulation correlating incoming requests across every server in real time under a single client ID, evaluating the entire multi-step sequence at once completely neutralized the evasion technique, dropping the attack success rate from 100.0% to 0.0% — demonstrating that cross-server state tracking, while not yet part of our deployed system, would close this gap entirely if implemented. An 83-case correctness suite and a 40-case synthetic attack benchmark validate the implementation, achieving an 80.0% overall mitigation coverage boundary.

**Index Terms**—Model Context Protocol, prompt injection, session state, multi-turn attacks, protocol security, LLM agents, capability attestation

## I. INTRODUCTION

### A. Context

The Model Context Protocol (MCP), introduced by Anthropic in November 2024, provides a standardized JSON-RPC 2.0 interface through which AI clients communicate with

tool-serving backends, with over 5,000 community-developed servers within months of release [1]. By separating the AI client from tool implementations behind a well-defined protocol boundary, MCP enables composable, reusable tooling ecosystems—but the same separation creates a new trust boundary that the protocol does not itself secure.

### B. Problem

Prior work identifies structural protocol-level vulnerabilities in MCP: absent capability attestation, unauthenticated sampling, and implicit cross-server trust propagation [1]. Proposed frameworks such as ATTESTMCP reduce measured attack success from 52.8% to 12.4% by evaluating each protocol message independently against a fixed policy. This is sufficient for attacks that are detectable from a single request in isolation. It is not sufficient, by construction, for attacks where the malicious signal only exists in the relationship between requests.

Consider: an agent issues three consecutive, legitimate resources/read calls to inspect project files, followed by a standard sampling/createMessage call to summarize findings. Individually, every single JSON-RPC message is completely benign and adheres strictly to the protocol specification — there are no malicious payloads, forbidden SQL keywords, or invalid execution calls. However, when evaluated as a sequence, the accumulation of sensitive file contexts immediately preceding an outbound LLM sampling call forms a stateful exfiltration vector. Because a single-turn evaluator  $f(\text{request}, \text{config})$  assesses each payload in isolation, it is structurally incapable of recognizing that the final request completes an attack sequence.

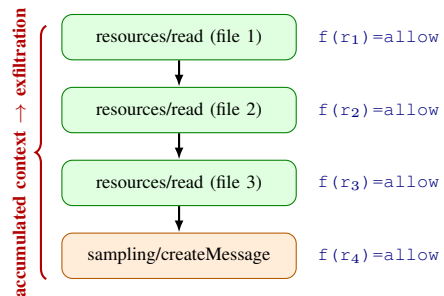


Fig. 1. Each request is individually approved by the single-turn evaluator  $f(\text{request}, \text{config})$ . The malicious signal exists only in the relationship between requests: three benign reads followed by a sampling call that, given the accumulated context, completes data exfiltration.

### C. Motivation

Empirical studies of agentic coding editors show attack success rates reaching 84% when external resources are poisoned [5], and concurrent work catalogues over 40 distinct MCP-specific attack techniques [2], [3]. Several of these are sequential by nature: each step is individually unremarkable, yet the combination causes harm.

Take a `rug-pull` update: a tool starts out looking completely normal and gets approved, then later gets quietly swapped for a malicious version. Neither the original nor the updated version looks suspicious on its own — the problem only shows up when the two are compared over time. Similarly `data-staging exfiltration`: instead of one big suspicious read, the attacker breaks the sensitive data into several small reads, each too small to trigger any alarm, then sends it all out in one final request. And `gradual privilege escalation` works the same way — each new permission requested is just a small step up from the last, so no single request ever looks like too much, even though the end result is way more access than the agent should have.

In every case, a single-turn evaluator only ever looks at one request at a time, with nothing to compare it to. So it simply can't see attacks that only become visible when you look at the sequence of requests, not any one of them alone. If single-turn evaluation is blind to an entire attack class by construction, then no amount of refinement to per-request rules closes that gap. This isn't just about writing better rules — it is a limit on what per-request checking can ever catch. For example, imagine an agent that reads three project files one at a time, then sends a summary request to an LLM. Each of those four requests looks completely normal on its own, so no matter how carefully you tune the rules a single-request checker uses, it will approve all four — the checker simply never gets to see them together. Adding more rules or stricter thresholds doesn't fix this, because the problem isn't that the rules are too loose; it's that the checker only ever looks at one request at a time. This is the gap we address.

### D. Contributions

This paper makes one central claim: detecting multi-step MCP attack sequences requires evaluators with access to session-level call history, and single-turn evaluators cannot detect this attack class regardless of rule sophistication. We validate this claim with three lines of evidence:

- **Necessity Argument:** We give a formal argument that any policy evaluator of the form  $f(\text{request}, \text{config})$  cannot detect attack sequences constructed such that no individual request satisfies a blocking predicate in isolation.
- **Empirical Validation:** Across 10 sequence-dependent benchmark cases, single-turn evaluation detects 0.0%; our session-aware extension detects 100.0%.
- **Detection/False-Positive Characterization:** We sweep sliding-window sizes  $w \in \{2, 3, 4, 5, 7, 10\}$  and measure how the true detection rate and false-positive rate change as the window grows, showing that multi-turn false-

positive rates escalate from 28.6% ( $w = 2$ ) to 100.0% ( $w = 5$ ) under adversarial configurations.

Beyond validating this claim, the paper makes two additional contributions:

- **Structural Boundaries:** We identify what independent, per-server session tracking can and cannot catch: it successfully detects multi-turn attacks built up within a single connection, but attacks split across multiple servers fall outside what any one server's session history can see. While protocol-layer session history tracking successfully isolate multi-turn state-building exploits originating from a single source connection, we identify cross-server multi-turn attacks as a limitation of localized, per-server session tracking, and map out this evasion case in our evaluation for future work.
- **Operationalization:** We build and evaluate MCP-Shield, the first open-source proxy operationalizing these protocol defenses, validated via an 83-case correctness suite and a 40-case synthetic attack benchmark (Table II) yielding 80.0% mitigation coverage.

## II. BACKGROUND AND RELATED WORK

### A. MCP Architecture

MCP defines a client-server architecture with three roles: Host, Client, and Server, communicating via JSON-RPC 2.0 over stdio or HTTP/SSE. The protocol defines three capability types: Resources, Tools, and Sampling (server-initiated LLM completions via `sampling/createMessage`). Sampling is critical from a security perspective for a simple reason: when a server sends a prompt to the host via `sampling/createMessage`, the host has no way to tell that prompt apart from something the actual user typed. In every host tested by [1], there is no UI indicator, warning, or separate trust level that distinguishes a server-injected prompt from legitimate user input — the LLM processes both identically. This means a compromised or malicious server can inject instructions directly into the model's context with the same authority as the person actually using the agent.

### B. Protocol-Level Vulnerabilities and ATTESTMCP

Maloyan and Namiot conduct the first rigorous security analysis of MCP, identifying capability escalation, unauthenticated sampling, and implicit cross-server trust propagation as structural weaknesses, validated across 847 attack scenarios within the PROTOAMP framework [1]. Among the vulnerability classes they identify, two are directly relevant to our threat model: `indirect injection` through resources and tool outputs, where malicious content planted in a resource or tool response is later consumed by the host as if it were trusted context, and `unauthorized sampling escalation`, where a server-initiated `sampling/createMessage` call is used to trigger LLM behavior the user never requested. The implicit cross-server trust propagation identified in [1] is what enables a related failure mode we study directly in this paper: an attack sequence split across multiple independent servers so that

no single server observes the complete pattern, which we refer to throughout as a cross-server multi-turn attack (§III, §V-D). ATTESTMCP adds capability attestation, HMAC-SHA256 authentication, origin tagging, and replay protection — all evaluated per-request. We adopt ATTESTMCP as our protocol-layer foundation and identify session-state evaluation as the dimension it does not cover.

### C. Empirical Attack Studies

Liu et al. demonstrate attack success rates reaching 84.0% on agentic coding editors using AIShellJack, a framework of 314 payloads spanning 70 MITRE ATT&CK techniques [5]. Maloyan and Namiot extend this to 42 exploit vectors in MCP-based coding assistants, including a class of tool response manipulation attacks — most notably shadow registration, where a malicious server silently registers a tool under a name already trusted by the host, and namespace squatting, where a server claims a tool name likely to collide with a legitimate one — both of which bypass detection because the manipulated tool listing itself is a single, protocol-valid message [2]. Neither study’s attack corpus is exclusively single-turn; several documented techniques (credential staging followed by exfiltration, trust-building before a malicious action) are inherently sequential, motivating our focus.

### D. Defense Placement and Persistent Threats

Rostamzadeh et al. introduce MCP-DPT, a defense-placement taxonomy organizing attacks by architectural enforcement layer [3]. We adopt their placement framing: MCP-Shield covers protocol interception and data-processing layers; MCP-Box covers execution isolation. Maloyan and Namiot separately address persistent sleeper channels—attacks that plant artifacts and trigger across session or process boundaries—proposing Provenance Gates, a cross-session architecture [4]. Our session-state contribution is narrower and complementary: we address attacks detectable within a session’s call history, explicitly leaving cross-session and cross-server correlation as the open problem that Provenance Gates address.

## III. THREAT MODEL AND THE NECESSITY ARGUMENT

We adopt the threat model of Maloyan and Namiot [1] and extend it formally to multi-turn attacks.

- *Attacker Capabilities:* The adversary controls or has compromised one MCP server in a multi-server deployment, and can inject content into data sources (web pages, documents, database records) that servers retrieve. The adversary has black-box access only—they cannot modify LLM weights or host application code.
- *Attacker Goals:* Hijacking agent behavior, exfiltrating data accessed through legitimate tool calls, and persisting across sessions via injected artifacts.
- *Attack Surface:* Four classes corresponding to the vulnerability classes of [1], [2], introduced in §II: (1) indirect injection through resources and tool outputs; (2) tool response manipulation, including shadow registration and

namespace squatting; (3) cross-server multi-turn attacks through context contamination; and (4) unauthorized sampling escalation.

### A. Formal Necessity Argument

We define a *single-turn evaluator* as any function

$$f : (\text{request}, \text{config}) \rightarrow \{\text{allow}, \text{block}\}$$

that depends only on the current request and static configuration.

**Definition:** We say a sequence  $r_1, r_2, \dots, r_n$  constitutes a successful attack if its execution achieves at least one attacker goal defined in our threat model — data exfiltration, unauthorized behavior hijacking, or persistence of a malicious artifact across sessions — where that outcome would not occur had any  $r_i$  been blocked.

**Claim:** There exist attack sequences  $r_1, r_2, \dots, r_n$  such that for every single-turn evaluator  $f$  and every prefix  $i < n$ ,  $f(r_i, \text{config}) = \text{allow}$ , yet the sequence collectively constitutes a successful attack.

**Construction:** Let the attack consist of  $n - 1$  requests  $r_1, r_2, \dots, r_{n-1}$  such that  $f(r_i, \text{config}) = \text{allow}$  for every single-turn evaluator  $f$  and every  $i \leq n - 1$  (e.g., reading resources a legitimate agent would plausibly read), followed by a final request  $r_n$  for which  $f(r_n, \text{config}) = \text{allow}$  as well (e.g., a sampling call), but whose effect — given the accumulated context from  $r_1, r_2, \dots, r_{n-1}$  — completes data exfiltration or behavior hijacking. By construction, no per-request predicate over  $r_i$  alone can distinguish this sequence from a benign one, since each  $r_i$  is drawn from the same distribution as benign traffic. Therefore  $f$  allows every  $r_i$ , including  $r_n$ , and the attack succeeds.

**Consequence:** By the claim above, for every single-turn evaluator  $f$  — regardless of how its per-request predicate is chosen or tuned — there exists an attack sequence that  $f$  allows in full. Since this holds for arbitrary  $f$ , no single-turn evaluator can detect this attack class; detection therefore requires an evaluator of the form  $g(\text{history}, \text{request}, \text{config})$  with access to prior call history within the session. This is a structural limitation of the single-turn evaluator’s domain (it is a function of the current request alone) rather than of any particular rule set — no choice of  $f$ , however sophisticated, escapes the claim, since the claim quantifies over all  $f$ .

We illustrate this necessity result concretely in §V-B, where 10 attack sequences constructed to satisfy this construction are, as the argument predicts, missed entirely by single-turn evaluation and fully detected once session history is available.

## IV. DESIGN AND IMPLEMENTATION

MCP-Shield instantiates  $g(\text{history}, \text{request}, \text{config})$  as a sequential pipeline. Stages 1, 2, 3, 4, and 5 realize the per-request component (equivalent to ATTESTMCP); Stage 1.5 realizes the history-dependent component.

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**Algorithm 1** MCP-Shield Evaluation Pipeline

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**Require:** request  $r$ , session history  $H$ , config  $C$

**Ensure:** decision  $\in \{\text{allow}, \text{block}\}$

```
1: Stage 1 (HMAC-SHA256 Verification)
2: if HMAC invalid or nonce replayed then
3:   return BLOCK
4: end if

5: Stage 1.5 (Sequence Detection)
6:  $H \leftarrow H \cup \{r\}$   $\triangleright$  append current request to history
7: if  $H$  matches an attack pattern within window  $w$  then
8:   return BLOCK
9: end if

10: Stage 2 (Capability Attestation)
11: if capability certificate invalid or unverified then
12:   return BLOCK
13: end if

14: Stage 3 (Regex Argument Scanning)
15: if tool arguments match blacklist regex then
16:   return BLOCK
17: end if

18: Stage 4 (AST-Based Code Analysis)
19: if AST walk finds blocked import/call/attribute then
20:   return BLOCK
21: end if

22: Stage 5 (Tool Namespace Lock)
23: if tool not in allowed_tools whitelist then
24:   return BLOCK
25: end if
26: return ALLOW
```

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Stages 1, 2, 3, 4, 5 realize the per-request component (equivalent to AT-TESTMCP [1]); **Stage 1.5** realizes the history-dependent component and is the original contribution of this work.

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### A. MCP-Shield Pipeline

1) *Stage 1: HMAC-SHA256 Verification:* Each HTTP/SSE request carries an `mcpsec` header with `server_id`, `timestamp`, `nonce`, and an HMAC-SHA256 signature over the request body using a pre-shared key. Verification uses `hmac.compare_digest` (constant-time). A sliding nonce window (30 s TTL) rejects replays.

2) *Stage 1.5: Multi-Turn Sequence Detection:* A `SessionStore` maintains per-server `SessionState` objects with call history (configurable TTL, default 1800 s). Before per-request evaluation, the current request is appended to history and checked against configured sequence rules of two types: pattern matching (ordered method/tool sequences within a sliding window) and rate limiting (call frequency within a window, independent of order). Pattern matching addresses the order-dependent attack signal present in all 10 sequence-dependent benchmark cases (Table VIII). Rate

limiting is included as a complementary, order-agnostic safeguard against high-frequency abuse (e.g., resource exhaustion or repeated probing) that would evade a purely pattern-based check if an attacker varied call order while preserving volume; our current benchmark does not yet include a case isolating this mechanism, which we note as a gap for future benchmark expansion. This stage directly realizes the history-dependent evaluator  $g$  required by §III-A and is the original contribution of this work.

3) *Stage 2: Capability Attestation:* Servers present a signed X.509 capability certificate during `initialize`. MCP-Shield verifies the signature against a trusted root CA, the validity window, and that the subject CN/SAN matches the claimed `server_id`. Verified capabilities gate subsequent method calls.

4) *Stage 3: Regex Argument Scanning:* Tool arguments are recursively scanned against a configurable regex blacklist (`rm\s+--rf, /etc/passwd, curl.*bash`, etc.).

5) *Stage 4: AST-Based Code Payload Analysis:* Code-bearing tool arguments are parsed via Python’s `ast` module and walked for blocked imports (`os`, `subprocess`, `socket`), blocked calls (`eval`, `exec`, `getattr`), and blocked attribute access (`popen`, `system`), catching obfuscation regex cannot.

6) *Stage 5: Tool Namespace Lock:* `tools/list` responses are filtered against a per-server `allowed_tools` whitelist, preventing shadow tool registration; non-whitelisted tool calls are rejected at `tools/call`.

7) *Output Sanitization:* Tool responses, `resources/read`, and `prompts/get` responses are passed through a two-pass sanitizer (line-start role-prefix matching plus full-text substring matching for injection phrases) before reaching the LLM context.

### B. MCP-Box: Ephemeral Execution Sandbox

MCP-Box executes Shield-cleared code payloads in one-shot Docker containers (`python:3.11-alpine`, `network_mode: none`, 128 MB memory cap, read-only filesystem, non-root user, 2-second `asyncio.wait_for` watchdog). Containers are force-removed in a `finally` block; label-based pruning on startup removes orphans from previous runs. A subprocess-based mock fallback supports unit testing without Docker.

## V. EVALUATION

### A. Correctness Suite

We validate individual security layers via an 83-case test suite across 12 modules (Table I): schema validation, certificate handling, database concurrency, sandbox isolation, regex and AST policy enforcement, namespace filtering, HMAC verification, attestation, session-state persistence/expiry, and end-to-end flow against live mock servers.

### B. Validating the Necessity Argument

We isolate the subset of our 40-case benchmark that satisfies the construction in §III-A: cases where every individual



TABLE I  
CORRECTNESS SUITE SUMMARY

| Module                   | Cases     | Coverage                             |
|--------------------------|-----------|--------------------------------------|
| test_schemas             | 20        | JSON-RPC validation, Pydantic models |
| test_database            | 3         | WAL logger, concurrency, faults      |
| test_certs               | 8         | CA fixture, expiry, tampering        |
| test_box_isolated        | 6         | Exec, timeout, network, OOM, FS      |
| test_policy_regex        | 8         | Blacklist patterns                   |
| test_policy_ast          | 7         | Import, call, attribute, obfuscation |
| test_namespace_sanitizer | 7         | Namespace lock, sanitizer            |
| test_hmac                | 4         | Valid, bad key, replay, expiry       |
| test_attestation         | 4         | Valid, expired, wrong ID             |
| test_engine              | 3         | Stage ordering                       |
| test_session_state       | 5         | Persistence, sequence, expiry        |
| test_end_to_end          | 5         | E1-E5 live mock servers              |
| test_stdio_proxy         | 3         | Pass-through, block, sanitize        |
| <b>Total</b>             | <b>83</b> | All pass                             |

request is indistinguishable from benign traffic under the Stage 2–5 (and sanitizer) predicates, and only the sequence is malicious. Ten cases meet this criterion. Table IX provides the full protocol-level specification of each case, including the exact sequence of the RPC method and the stateful exploitation mechanism. Appendix A walks through two of these cases — MPS-020 and MPS-026 — at the level of individual JSON-RPC messages, illustrating concretely how each request is benign in isolation and only becomes malicious as part of the accumulated sequence. To establish an ATTESTMCP baseline for comparison, we evaluate each case under two configurations: a baseline configuration with Stage 1.5 disabled — a fresh `SessionState` per request, equivalent to pure ATTESTMCP with no session history — and our full configuration with Stage 1.5 enabled (Table VIII). This lets us isolate the detection improvement that is specifically attributable to session-aware sequence tracking, rather than to any other stage in the pipeline.

The remaining 30 cases are single-turn attacks — detectable from a single request in isolation via Stages 2–5 (capability attestation, regex/AST scanning, namespace locking) or the output sanitizer — and are evaluated separately as part of the full 40-case benchmark in §V-E (Table II), where we report coverage across all four attack classes rather than the sequence-dependent subset alone.

TABLE II  
SYNTHETIC BENCHMARK COVERAGE (40 CASES)

| Attack Class             | Cases     | Blocked           | Missed           | Seq.-dep. |
|--------------------------|-----------|-------------------|------------------|-----------|
| V1: Indirect Injection   | 10        | 6                 | 4                | 0         |
| V2: Tool Response Manip. | 10        | 8                 | 2                | 1         |
| V3: Cross-Server Prop.   | 10        | 8                 | 2                | 4         |
| V4: Sampling Injection   | 10        | 10                | 0                | 5         |
| <b>Total</b>             | <b>40</b> | <b>32 (80.0%)</b> | <b>8 (20.0%)</b> | <b>10</b> |

<sup>†</sup> This uses MCP-Shield’s default configuration: a session timeout of 1800s and a cap of 100 calls per session, but bypasses the rule for a window size of 3 to scan full session call history with no fixed sliding-window bound.

### C. Detection / False-Positive Tradeoff Across Window Size

The necessity argument shows session state is required; it does not determine how much history ( $w$ ) is needed,

or what false-positive cost a given window size imposes on benign workloads. We sweep sliding-window size  $w \in \{2, 3, 4, 5, 7, 10\}$  across the 10 sequence-dependent cases from Table VIII alongside a dedicated multi-turn false-positive benchmark of 14 legitimate workflows (Table III), evaluated separately from the attack benchmark. This benchmark includes four baseline workflow types — legitimate summarization, data pipelines, file-copy operations, and multi-turn conversations — each tested both standalone and with a trailing extra step (8 cases), plus six workflows specifically constructed to closely resemble attack patterns at the protocol level: a rapid burst of benign calls, a benign sequence with an attack pattern embedded partway through, a deferred subsequence match, a stuttering read pattern, a partially abandoned workflow, and a legitimate read-modify-write flow structurally identical to a data-staging attack. This corpus was deliberately constructed to probe the detection boundary and is not representative of general benign traffic.

The evaluation shows a trade-off sharper than a smooth curve, not a gradual one: true detection rate saturates at 100.0% by  $w = 4$ , but the false-positive rate has already reached 92.86% at that same window, and climbs to 100.0% by  $w = 5$  with no further detection gain. The false-positive rate climbs from 92.86% at  $w = 4$  to 100.0% at  $w = 5$  as one additional benign case — otherwise correctly allowed at smaller windows — crosses the pattern-match threshold once enough history accumulates.

There is no window size in this sweep at which the system achieves complete attack detection without also flagging nearly every case in this adversarial benign corpus. Four of the fourteen cases (FPMT003, FPMT009, FPMT010, FPMT014) are flagged at every tested window size, including  $w = 2$ , indicating these specific legitimate patterns are structurally indistinguishable from an attack regardless of  $w$ ; resolving these requires more specific sequence patterns (§IV-A2), not a change in window size.

TABLE III  
MULTI-TURN FALSE-POSITIVE BENCHMARK SUMMARY

| Case ID | Category           | Description                           |
|---------|--------------------|---------------------------------------|
| FPMT001 | Baseline           | legitimate summarization flow         |
| FPMT002 | Baseline           | legitimate data pipeline              |
| FPMT003 | Baseline           | legitimate file-copy operation        |
| FPMT004 | Baseline           | legitimate multi-turn conversation    |
| FPMT005 | Trailing           | summarization + trailing step         |
| FPMT006 | Trailing           | copy operation + trailing step        |
| FPMT007 | Trailing           | pipeline + trailing step              |
| FPMT008 | Trailing           | conversation + trailing step          |
| FPMT009 | Adversarial-shaped | rapid burst of benign calls           |
| FPMT010 | Adversarial-shaped | attack pattern embedded mid-sequence  |
| FPMT011 | Adversarial-shaped | deferred subsequence match            |
| FPMT012 | Adversarial-shaped | stuttering read pattern               |
| FPMT013 | Adversarial-shaped | partially abandoned workflow          |
| FPMT014 | Adversarial-shaped | read-modify-write (data-staging-like) |

For operators selecting  $w$  in a general deployment, we recommend tuning to the longest ordinary multi-step workflow the deployment is expected to produce — not to the longest attack sequence — since Table IV shows detection saturates

well below the point where false positives become severe, and increasing  $w$  further only adds false-positive cost with no additional detection benefit. Cases structurally indistinguishable from an attack at any window size should instead be addressed through more specific pattern rules rather than window tuning. Note that Table II’s evaluation uses an unbounded window to demonstrate that session-aware detection is possible at all, consistent with the necessity argument in §III-A; per the results above, this is a research configuration, not a recommended production setting, since an unbounded window maximizes both false-positive rate and per-request scan cost as session length grows.

TABLE IV  
OPERATIONAL SEQUENCE-RULE WINDOW SIZE EVALUATION

| Horizon Boundary ( $w$ ) | True Sequence Detection Rate | Multi-Turn False-Positive Rate |
|--------------------------|------------------------------|--------------------------------|
| $w = 2$                  | 20.00%                       | 28.57%                         |
| $w = 3$                  | 90.00%                       | 64.29%                         |
| $w = 4$                  | 100.00%                      | 92.86%                         |
| $w = 5$                  | 100.00%                      | 100.0%                         |
| $w = 7$                  | 100.00%                      | 100.0%                         |
| $w = 10$                 | 100.00%                      | 100.0%                         |

The reading of “100.0% / 100.0%” at  $w \geq 5$  does not mean the system blocks all traffic indiscriminately; it means every attack case is still correctly caught, while every benign case in this deliberately adversarial corpus is also incorrectly flagged. The two rates are not directly comparable to a single “percent of traffic blocked” figure.

#### D. Sufficiency of Local Session State

We isolate cross-server propagation attacks (MPS-026, MPS-027, MPS-030) to assess whether localized tracking within an isolated server deployment (Condition A) is sufficient to mitigate stateful vulnerabilities (Table V). In a real deployment, MCP-Shield tracks session history separately for each backend server, since pooling session data across servers could leak one tenant’s activity into another’s session view in a shared, multi-tenant deployment. To evaluate what would be gained if this constraint were lifted, we built a standalone simulation modeling a unified tracking configuration (Condition B), in which session history from every server is pooled under a single client identifier, so a sequence split across Server A, B, and C is visible as one continuous history rather than three isolated ones. This lets us measure the security upper bound achievable if cross-server session correlation were implemented.

Table V isolates these three cross-server cases; the related restart-persistence case, MPS-029, is discussed separately in §V-H, as it tests a different mechanism (persistence under restart) rather than cross-server tracking topology.

TABLE V  
CROSS-SERVER MULTI-TURN ATTACK ABLATION SUMMARY

| Case ID    | Behavioral Description      | Condition A (Isolated) | Condition B (Unified) |
|------------|-----------------------------|------------------------|-----------------------|
| MPS-026    | Data staging via sampling   | missed                 | BLOCKED               |
| MPS-027    | Cascade attack              | missed                 | BLOCKED               |
| MPS-030    | Below window size (14-read) | missed                 | BLOCKED               |
| <b>ASR</b> | <b>Attack Success Rate</b>  | <b>100.00%</b>         | <b>0.00%</b>          |

#### E. Full Benchmark Coverage

Beyond the sequence-dependent cases, we evaluate the complete 40-case synthetic benchmark (Table II).

The 8 missed cases represent known structural boundaries: 4 comprise sophisticated sanitizer encoding obfuscations (Unicode substitution, zero-width space inclusions, base64-splits) and 4 map to semantic/architectural gaps.

Table VI attributes each blocked case to the specific pipeline stage responsible for catching it. Without Stage 1.5, the remaining stages (capability attestation, regex/AST scanning, namespace locking, and output sanitization) collectively block 22 of the 40 benchmark cases — equivalent to the ATTESTMCP-only baseline. Adding Stage 1.5 raises this to the full 32 of 40 reported in Table II. This confirms that all 10 cases uniquely closed by our contribution are exactly the sequence-dependent cases identified in §V-B, with zero overlap against cases already caught by per-request stages.

TABLE VI  
STAGE ATTRIBUTION FOR BLOCKED CASES

| Defense Stage             | V1       | V2       | V3       | V4        | Total     |
|---------------------------|----------|----------|----------|-----------|-----------|
| Stage 1.5 (Session State) | 0        | 1        | 4        | 5         | 10        |
| L1 (Attestation / HMAC)   | 0        | 0        | 1        | 1         | 2         |
| Regex / AST Scanners      | 0        | 2        | 2        | 0         | 4         |
| Output Sanitizer          | 6        | 5        | 1        | 4         | 16        |
| <b>Blocked Total</b>      | <b>6</b> | <b>8</b> | <b>8</b> | <b>10</b> | <b>32</b> |
| <b>Missed (no stage)</b>  | <b>4</b> | <b>2</b> | <b>2</b> | <b>0</b>  | <b>8</b>  |

#### F. Performance Overhead

Per-request latency overhead is measured over 1,000 requests with warm SQLite connections (Table VII).

Total protocol-layer overhead introduces a P50 median latency of just 0.153 ms. The Stage 1.5 history evaluation accounts for a negligible 0.008 ms, proving that stateful protocol defenses do not impose a heavy deployment tax.

TABLE VII  
MCP-SHIELD PER-REQUEST LATENCY OVERHEAD

| Stage                         | P50 (ms)     | P95 (ms)     |
|-------------------------------|--------------|--------------|
| HMAC verification             | 0.047        | 0.083        |
| Sequence check (Stage 1.5)    | 0.005        | 0.006        |
| Attestation (cert cached)     | 0.000        | 0.000        |
| Regex scan                    | 0.004        | 0.005        |
| AST parse + walk              | 0.058        | 0.067        |
| Namespace lock                | 0.000        | 0.000        |
| Output sanitizer              | 0.039        | 0.051        |
| <b>Total (no code exec)</b>   | <b>0.153</b> | <b>0.212</b> |
| <b>MCP-Box Container exec</b> | <b>0.008</b> | <b>0.008</b> |

#### G. Stateless Structural Rule Sensitivity and False-Positive Boundaries

This benchmark corpus was deliberately adversarial, not representative of general web traffic: we constructed a ded-

TABLE VIII  
SINGLE-TURN VS. SESSION-AWARE DETECTION (SEQUENCE-DEPENDENT SUBSET)

| Case ID               | Sequence Trigger                    | Single-Turn Evaluator $f$ | Session-Aware Proxy $g$ |
|-----------------------|-------------------------------------|---------------------------|-------------------------|
| MT-01 (MPS-020)       | gradual escalation                  | Missed                    | BLOCKED                 |
| MT-02 (MPS-026)       | data staging via sampling           | Missed                    | BLOCKED                 |
| MT-03 (MPS-027)       | cascade attack                      | Missed                    | BLOCKED                 |
| MT-04 (MPS-029)       | cross-session restart               | Missed                    | BLOCKED                 |
| MT-05 (MPS-030)       | below window size (14-read)*        | Missed                    | BLOCKED                 |
| MT-06 (MPS-033)       | rate-limit probe maxTokens          | Missed                    | BLOCKED                 |
| MT-07 (MPS-034)       | sequential sampling context buildup | Missed                    | BLOCKED                 |
| MT-08 (MPS-035)       | resource-to-sampling exfiltration   | Missed                    | BLOCKED                 |
| MT-09 (MPS-037)       | sampling after tool sequence        | Missed                    | BLOCKED                 |
| MT-10 (MPS-039)       | slow-burn sampling*                 | Missed                    | BLOCKED                 |
| <b>Detection rate</b> |                                     | <b>0.0%</b>               | <b>100.0%</b>           |

\*Session-Aware Proxy  $g$  is evaluated here with Stage 1.5 scanning the full session history (no fixed window bound). The windowed sensitivity analysis in Table IV separately constrains detection to a fixed window  $w$ .

TABLE IX  
PROTOCOL-LAYER SEQUENTIAL ANATOMY OF EVALUATED MULTI-TURN ATTACK TRACES

| Case ID | Behavioral Profile      | Trace Length ( $N$ ) | Protocol Method Sequence                                   | Stateful Exploitation Mechanism                                                                                                                                                              |
|---------|-------------------------|----------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MPS-020 | Gradual Escalation      | 3                    | resources/read<br>resources/read<br>sampling/createMessage | → Builds context across multiple reads, then executes an unauthorized model sampling call to exfiltrate data.                                                                                |
| MPS-026 | Data Staging            | 4                    | 3× resources/read<br>sampling/createMessage                | → Reads a sensitive resource across three consecutive turns before initiating an unauthorized sampling call to exfiltrate the accumulated context.                                           |
| MPS-027 | Cascade Attack          | 3                    | 2× resources/read<br>sampling/createMessage                | → Reads a trigger resource across two turns on a compromised server, transfers accumulated context to an outbound model channel."                                                            |
| MPS-029 | Cross-Session Restart   | 2                    | resources/read → (restart)<br>sampling/createMessage       | → Restarts gateway mid-sequence by clearing session state between the read and sampling phases. SQLite backing store remakes history on reconnection, sliding window spans restart boundary. |
| MPS-030 | Below-Window Read       | 15                   | 14× resources/read<br>sampling/createMessage               | → Executes a low-and-slow read sequence exactly matching or exceeding the proxy sliding window barrier ( $w$ ) to hide the final action.                                                     |
| MPS-033 | Rate-Limit Probe        | 3                    | 2× resources/read<br>sampling/createMessage                | → Executes low-privilege file reads before initiating a sampling request with excessively high token allocation (maxTokens=32000) to exhaust resources.                                      |
| MPS-034 | Sequential Sampling     | 3                    | 3× sampling/createMessage                                  | 3 consecutive sampling calls build injected context to bypass single-request anomaly detection and hijack persona constraints.                                                               |
| MPS-035 | Resource Hijacking      | 3                    | 2× resources/read<br>sampling/createMessage                | → Server reads a sensitive local resource and immediately uses that context to initiate an unauthorized sampling escalation.                                                                 |
| MPS-037 | Sampling after Sequence | 4                    | 3× tools/call<br>sampling/createMessage                    | → Executes 3 distinct tool calls (get_data, format_data, analyze) to stage data before initiating a sampling call to bypass sequence rules.                                                  |
| MPS-039 | Slow-Burn Sampling      | 11                   | 10× resources/read<br>sampling/createMessage               | → Executes 10 preceding clean file read calls to exhaust standard sequence tracking boundaries and launches the final malicious sampling request.                                            |

icated false-positive benchmark of 24 complex, single-turn benign development workloads (Table X), evaluated separately from the 40-case attack benchmark in §V-E — including ORM serializers utilizing dynamic reflection via getattr, multi-threaded data pipelines, and security hardening documentation referencing /etc/shadow. Under these conditions, the stateless rules engines produced a 100.0% false-positive rate, confirming that static signatures tight enough to catch single-

turn exploits inevitably flag benign code with identical syntax. Auxiliary tests run against the 24 single-turn samples confirm this trade-off: softening the strictness of the per-request regex and AST filters directly drops false positives, yielding a 75.0% baseline compliance rate.

This result exposes two distinct failure modes, each with a different cause and a different fix. The first is single-turn false positives in Stages 3–4, caused by AST and regex

rules that match on surface form rather than intent. For example, the AST scanner flags a legitimate ORM serializer for calling `getattr` during routine field access (FP003), and a graceful-shutdown handler for registering a `SIGTERM` signal handler (FP018) — both benign uses of blocklisted calls with no malicious intent. The output sanitizer similarly flags a security hardening guide that mentions `/etc/shadow` in a protective, documentary context (FP022), and a `systemd` unit file whose `system:` field is a configuration key rather than an injected role-prefix (FP015). In each case, the filter correctly identifies a blocklisted token but has no mechanism to distinguish a legitimate developer context from an actual attack payload — a limitation inherent to stateless, per-request analysis and independent of any sequence-level configuration. The second is multi-turn false positives (Table IV) attributable to Stage 1.5 sequence rules, which operate at the protocol layer and are intentionally blind to semantic context: a developer legitimately performing repeated resource reads followed by a sampling call is, at the protocol level, indistinguishable from an exfiltration sequence. Differentiating intent at the sequence level requires either semantic analysis or operator-supplied context. Both are outside the scope of a protocol-layer defense and remain open areas for future work.

TABLE X  
FALSE-POSITIVE BENCHMARK SUMMARY (24 CASES)

| Case ID | Category  | Trigger                                         |
|---------|-----------|-------------------------------------------------|
| FP001   | Sanitizer | "you are now" in deployment status              |
| FP002   | Sanitizer | "user message:" in chat log export              |
| FP003   | AST       | <code>getattr</code> in ORM serializer          |
| FP004   | AST       | threading in data pipeline                      |
| FP005   | AST       | <code>urllib.parse</code> for URL normalization |
| FP006   | Sequence  | read-then-write in JSON formatter               |
| FP007   | Sanitizer | "new instructions:" as CI log header            |
| FP008   | Sanitizer | "disregard your" in cache message               |
| FP009   | Session   | 101-call batch indexer                          |
| FP010   | Sanitizer | tool-listing help text                          |
| FP011   | Sequence  | 6-step ETL pipeline                             |
| FP012   | AST       | <code>setattr</code> in config loader           |
| FP013   | Sanitizer | "user" key in YAML config                       |
| FP014   | Sanitizer | "assistant" variable in source                  |
| FP015   | Sanitizer | "system:" key in systemd unit                   |
| FP016   | AST       | <code>app.run()</code> in Flask startup         |
| FP017   | AST       | <code>shutil.copy</code> for backup             |
| FP018   | AST       | <code>SIGTERM</code> handler registration       |
| FP019   | AST       | <code>re.compile</code> for validation          |
| FP020   | AST       | <code>delattr</code> stripping private fields   |
| FP021   | Sanitizer | "invoke server:" in AWS docs                    |
| FP022   | Sanitizer | <code>/etc/shadow</code> in hardening guide     |
| FP023   | Sequence  | legitimate 2-read summarization                 |
| FP024   | Sequence  | analytics pipeline name collision               |

These two failure modes have separate mitigations. Operators can reduce single-turn false positives by relaxing Stage 3–4 signatures without modifying Stage 1.5 — demonstrating that layering our session-aware contribution on top of AT-TESTMCP’s per-request stages preserves each layer’s independent tunability. Multi-turn false-positive rates depend on how sequence rules are configured, and should be tuned to an

operator’s risk tolerance: a deployment prioritizing availability may configure looser windows and fewer patterns, accepting a higher missed-attack rate, while a deployment prioritizing security may configure tighter rules at the cost of more false positives — both remain production configurations of the same system. The key architectural contribution is not that sequence rules eliminate false positives — some residual false-positive cost is inherent to any sequence-based check — but that Stage 1.5 closes the false-negative gap left by single-turn evaluation (§III-A), while remaining independently configurable from Stages 3–4. This lets operators tune each stage’s threshold separately according to their risk tolerance, rather than being forced to trade off detection and false-positive cost through a single shared parameter.

#### H. Adversarial Analysis of Process-Restart Evasion

We validate the system’s restart-resilience mechanism directly with MPS-029, a dedicated test case in which a single session accumulates two resource reads, the in-memory `SessionStore` is then cleared to simulate a cold gateway restart, and a final `sampling/createMessage` call is issued in a reconstructed session. We term the vulnerability this targets the memory-wipe evasion vector: if session history were held only in volatile process memory, an attacker could evade detection simply by forcing or waiting for a restart, silently erasing the call history Stage 1.5 depends on and defeating session tracking entirely without touching the sequence-detection logic itself. MCP-Shield closes this gap by persisting session history to a SQLite-backed store rather than memory alone; MPS-029 confirms this empirically, with the proxy reconstructing history from SQLite after the in-memory wipe and correctly blocking the attack.

A more surgical adversary might attempt to evade Stage 1.5 not by wiping the entire session store, but by timing a process crash or restart to land in a narrow window: after a malicious request has already been evaluated and allowed, but before its corresponding history entry has finished being written to SQLite. If this timing succeeds, that specific entry would be lost, and the reconstructed history on restart would be missing evidence of the attacker’s prior calls. However, MCP-Shield’s write dispatch is asynchronous rather than synchronous: each call is recorded in memory immediately, so the allow/block decision is never blocked on a database write, while the corresponding write is dispatched as a background task to the event loop. This means the attacker cannot precisely predict or control when the write completes relative to their crash, since the write is scheduled independently of the request/response cycle rather than gated behind it. MPS-029 does not test this narrow window directly — it simulates a coarse-grained full store clear rather than a precisely timed crash during an in-flight write — so a sufficiently well-timed attack against this async write remains a residual risk not directly evaluated in this work.

This SQLite-based design has one remaining limitation: it persists history to a single file on a single machine, so a production deployment running multiple parallel proxy work-



ers would still be vulnerable if a malicious sequence were split across two different workers, since each worker’s SQLite file only contains the requests that worker itself handled. Replacing SQLite with a shared, network-accessible backend such as Redis would extend the same restart-survival guarantee across every worker in the deployment, closing this remaining single-process dependency.

## VI. LIMITATIONS AND FUTURE WORK

- **Window-size selection:** The sharp tradeoff curve in Table IV indicates that selection of  $w$  remains highly environment-dependent; a uniform  $w$  does not dominate across varied deployment models.
- **Cross-server session correlation:** The deployed system (Condition A) achieves 0.0% detection on cross-server multi-turn attacks. As demonstrated by the simulated Condition B evaluation in §V-D, consolidating session history across separate upstream servers into a single unified tracking configuration closes this gap entirely, raising detection to 100.0%. However, moving this from a standalone simulation to a live multi-tenant deployment introduces two concrete costs.

First, correlating requests across multiple servers in real time requires each server to synchronously report activity to a shared session store, adding network round-trip latency to the request path that a purely local, per-server store avoids. Second, pooling session history across servers means a proxy operator would need visibility into every tenant’s activity across every server they use, which conflicts with the isolation guarantees a shared, multi-tenant deployment is expected to provide — a tenant using one server should not have their session data readable by whatever process manages another server’s history. Condition B quantifies the potential security gains of this approach, though resolving these latency and isolation trade-offs remains an objective for future engineering.

- **Session Identity Rotation and Server Pinning:** A structural boundary of the current deployment is its dependency on client-supplied transportation headers (`x-mcpsec-server-id`) to key persistent session history. An adversary who programmatically rotates this identifier on every subsequent request can deliberately spawn decoupled, orthogonal session streams, thereby fragmenting the historical sliding window and evading multi-turn detection.

The root cause of this vulnerability lies in the lack of cryptographic verification of the upstream server’s identity at the proxy ingress. To resolve this limitation in production environments, the tracking topology would benefit from moving beyond passive header parsing toward stronger server identity verification. One possible solution is strict, non-repudiable server identity pinning: future versions of MCP-Shield could enforce mutual TLS (mTLS) or hardware-backed Trusted Platform Module (TPM) identity attestation, binding the sliding-window telemetry token directly to the verified cryptographic

handshake of the upstream host. This would ensure state continuity even if transport-layer headers are intentionally obfuscated or rotated, though other approaches — such as signed session tokens or a trusted server identity registry — could plausibly address the same root cause and are not evaluated in this work.

- **Scope Boundaries:** This work addresses attacks detectable through protocol-layer session state and does not address several adjacent threat classes. Transport-layer attacks requiring TLS termination — e.g., an adversary who can intercept or terminate the TLS connection itself — operate below the JSON-RPC layer our proxy inspects, and would require transport-security hardening (certificate pinning, mTLS) rather than sequence-based detection. Kernel-level container escapes target the isolation boundary of MCP-Box’s execution sandbox rather than the protocol layer MCP-Shield secures; defending against these requires kernel-hardening and sandbox-escape mitigations orthogonal to session tracking. Social engineering attacks manipulate the human operator directly rather than the protocol, and no protocol-layer defense — stateful or otherwise — can observe or intervene on that channel. Attacks confined entirely within a single, legitimately certified server fall outside our threat model because our adversary model assumes the attacker controls or has compromised a server; a fully trusted, uncompromised server acting maliciously through legitimate certified capabilities is a capability-attestation problem, not a session-tracking one. Finally, sleeper channels requiring cross-session persistence — attacks that plant an artifact in one session and trigger it in a later, unrelated session — fall outside the single-session window our SessionStore tracks; Maloyan and Namiot’s Provenance Gates architecture [4] addresses this complementary problem directly, and we treat it as outside our scope rather than a limitation of our specific design.

## VII. CONCLUSION

This paper began with a simple observation: a policy evaluator that only ever sees one request at a time cannot, by construction, detect an attack whose signal lives in the relationship between requests. We formalized this as a necessity result, not a rule-coverage gap — no amount of tuning a single-turn evaluator closes it — and confirmed the result empirically, where session-aware tracking took detection on our sequence-dependent benchmark from 0.0% to 100.0% at a cost of just 0.008 ms per request. But the picture that emerged from the full evaluation is more nuanced than that headline number alone suggests. Session-state tracking is necessary, but it is not sufficient on its own: our window-size sweep showed a sharp trade-off between catching longer attack sequences and triggering false positives on legitimate development workflows, meaning  $w$  must be tuned per deployment rather than fixed universally. Our cross-server ablation showed that

localized, per-server tracking — the configuration we recommend as a default for multi-tenant deployments — misses attacks deliberately split across servers entirely, and that closing this gap requires trading away the very isolation that motivated the local design in the first place. And our false-positive analysis showed that stateless per-request filtering (Stages 3–4) and stateful sequence rules (Stage 1.5) fail in different ways, for different reasons, and need to be tuned independently rather than as a single knob.

Taken together, these results argue for a specific, narrower claim than add more session state and the problem is solved: session-level history is a necessary architectural primitive for MCP security, but it introduces its own configuration surface — window size, tracking topology, persistence guarantees — that a deployment must reason about explicitly rather than treat as solved by inclusion alone. We see MCP-Shield’s 80.0% coverage on the full synthetic benchmark, and the open cross-server and identity-pinning gaps documented in §VI, as evidence of how much of this surface remains to be explored, rather than as a finished answer. What we believe is settled is narrower and more durable: any MCP defense that evaluates requests one at a time, however well-tuned, will always be blind to this class of attack. Session state is not an optional enhancement to per-request checking — it is a different, necessary layer.

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## APPENDIX

To illustrate our necessity argument that stateful sequence inspection is required where stateless inspection fails, this appendix details two concrete attack traces evaluated in our synthetic benchmark suite (§V-B): **MPS-020 (Gradual Escalation)** and **MPS-026 (Data Staging via Sampling)**.

In both scenarios, every individual JSON-RPC request is entirely benign and perfectly compliant when viewed in isolation. A standard tool call, resource read, or sampling request conforms fully to the Model Context Protocol (MCP) specification. Detection and mitigation occur exclusively at

**Stage 1.5 (Sequence & Stateful Policy Check)** by evaluating the accumulated sliding window.

### A. Trace 1: MPS-020 (Gradual Escalation)

This sequence simulates an agent gradually escalating privileges or exfiltrating sensitive project files by staging harmless-looking reads before calling the sampling API.

#### 1) Turn 1 (Resource Read — Benign in isolation):

```
{"jsonrpc": "2.0", "method": "resources/read", "params": {"name": "file.txt"}, "id": 1}
```

*Action:* Logged in sliding window state. No single-request predicates violated.

#### 2) Turn 2 (Resource Read — Benign in isolation):

```
{"jsonrpc": "2.0", "method": "resources/read", "params": {"name": "file.txt"}, "id": 2}
```

*Action:* Appended to sliding window state. History length satisfies threshold.

#### 3) Turn 3 (Sampling Execution — Benign in isolation):

```
{"jsonrpc": "2.0", "method": "sampling/createMessage", "params": {}, "id": 3}
```

*Outcome:* **BLOCKED by Stage 1.5 (Sequence Check).** The policy engine identifies the high-risk pattern `[resources/read(project_notes.txt), resources/read(api_keys.txt)]` preceding the sampling request within the active sliding window, blocking exfiltration before the LLM context is compromised.

### B. Trace 2: MPS-026 (Data Staging via Sampling)

This sequence models a more distributed attack vector where an agent repeatedly stages small resource reads (or tool calls) across multiple turns to slowly stage data before pushing it to an external server via sampling.

#### 1) Turns 1–3 (Staged Resource Reads — Benign individually):

```
{"jsonrpc": "2.0", "method": "resources/read", "params": {"name": "secrets.txt"}, "id": 1}
{"jsonrpc": "2.0", "method": "resources/read", "params": {"name": "secrets.txt"}, "id": 2}
{"jsonrpc": "2.0", "method": "resources/read", "params": {"name": "secrets.txt"}, "id": 3}
```

*Action:* The unified session tracker records the repetitive read sequences over the temporal window without interrupting the client.

#### 2) Turn 4 (Sampling Execution):

```
{"jsonrpc": "2.0", "method": "sampling/createMessage", "params": {}, "id": 4}
```

*Outcome:* **BLOCKED by Stage 1.5 (Sequence Check).** Evaluated statically, `sampling/createMessage` is a fully legitimate operation. Evaluated statefully, the frequency and structure of the preceding reads trigger the sequence-anomaly rule, preventing the sampling payload from executing.

**Remark:** These traces illustrate why stateless evaluators fail. Because an evaluator  $f(\text{request})$  examines messages in isolation, it cannot distinguish between an attack sequence and an honest agent summarizing a repository.